The amplitude adapter for $\eta e^{\beta s \xi}$

Claude, with Daniel Murfet

2026-09-07

Abstract

The last module of the coefficient closure (Astra #7). From weighted-summable coefficient arrays x of ξ and y of η at radius ρ, the s-parametrised array $c_k(s) = e^{\beta s x_{00}}(y * E(\beta s))_k$, with E the exponential of $x - x_{00}\delta$, is the double series of $\eta(u, v)e^{\beta s\xi(u,v)}$ for $|u|, |v| \leq \rho$; each c_k is continuous in s and $|c_k(s)|\rho^{|k|} \leq Y e^{\beta s x_{00}} e^{|\beta s| X}$. Feeding this into `twoD_cutoff_analytic` with $L = x_{00} + X$, $D = 0$, $C_0 = Y$ gives `twoD_cutoff_amplitude`: the analytic $d = 2$ cutoff theorem for the chart amplitude of `thm:TaylorTree`, whose only analytic hypothesis is the weighted summability of the two coefficient arrays. Zero `sorry`/`axiom`.

1 The things formalised

```
noncomputable def dropConst (x :   ×   →  ) :   ×   →   :=
  fun k => if k = (0, 0) then 0 else x k
theorem dblSum_dropConst (  u v :  ) (x :   ×   →  ) (hx : WSummable   x) (hu : |u|    )
    (hv : |v|    ) : dblSum (dropConst x) u v = dblSum x u v - x (0, 0)
noncomputable def ampCoeff (  :  ) (x y :   ×   →  ) (k :   ×  ) (s :  ) :   :=
  Real.exp (  * s * x (0, 0)) * conv y (expCoeff (dropConst x) (  * s)) k
theorem ampCoeff_continuous (  :  ) (x y :   ×   →  ) (k :   ×  ) :
    Continuous (ampCoeff   x y k)
theorem ampCoeff_abs_le (    :  ) (h  : 0 <  ) (x y :   ×   →  )
    (hx : WSummable   x) (hy : WSummable   y) (k :   ×  ) (s :  ) :
    |ampCoeff   x y k s| *   ^ (k.1 + k.2)
       wnorm   y * Real.exp (  * s * x (0, 0))
        * Real.exp (|  * s| * wnorm   (dropConst x))
theorem dblSum_ampCoeff (    u v :  ) (h  : 0 <  ) (x y :   ×   →  )
    (hx : WSummable   x) (hy : WSummable   y) (hu : |u|    ) (hv : |v|    ) (s :  ) :
    dblSum (fun k => ampCoeff   x y k s) u v
      = dblSum y u v * Real.exp (  * s * dblSum x u v)
theorem anaAmp_ampCoeff (  b   :  ) (hb : 0 < b) (hb  : b <  ) (x y :   ×   →  )
    (hx : WSummable   x) (hy : WSummable   y) (u v s :  )
    (hu : u   Icc (0 :  ) b) (hv : v   Icc (0 :  ) b) :
    anaAmp (ampCoeff   x y) b u v s
      = dblSum y u v * Real.exp (  * s * dblSum x u v)
theorem twoD_cutoff_amplitude (  b p   :  ) (h  h  k  k  M  M  :  ) (h  : 0 <  )
    (hb : 0 < b)
    (hb  : b <  ) (hk  : 0 < k ) (hk  : 0 < k ) (hp  : ((h  :  ) + 1) / k  = p)
    (hp  : ((h  :  ) + 1) / k  = p)
    (hM  : ((M  :  ) - 1) / k  < (M  :  ) / k )
    (hM  : ((M  :  ) - 1) / k  < (M  :  ) / k )
    (x y :   ×   →  ) (hx : WSummable   x) (hy : WSummable   y) :
    (fun N :   => twoDAmp   b N h  h  k  k  (anaAmp (ampCoeff   x y) b)
```

```
  - reducedSum   b h  h  k  k  M  M  (anaFaceU (ampCoeff   x y) b)
      (anaFaceV (ampCoeff   x y) b) (fun i j s => ampCoeff   x y (i, j) s) N)
=O[atTop] fun N :   =>
  N ^ (-(p + min ((M  :  ) / k ) ((M  :  ) / k ))) * (1 + Real.log N)
```

2 Mathematics

Everything is assembly of units 117 and 118. Removing the constant term preserves weighted summability termwise; the pointwise identity $x = \mathrm{dropConst}\, x + x_{00}\delta$ and additivity of absolutely convergent double series give $\mathrm{ev}(\mathrm{dropConst}\, x) = \xi - x_{00}$. The weighted norm of $c(s)$ is $e^{\beta s x_{00}} N_\rho(y * E(\beta s)) \le e^{\beta s x_{00}} N_\rho(y) e^{|\beta s| X}$ by the closure and the exponential bound, and a single weighted term is at most the weighted norm. Evaluation is multiplicativity followed by the exponential identity and $e^{\beta s x_{00}} e^{\beta s(\xi - x_{00})} = e^{\beta s \xi}$. For $s \ge 0$, $\beta > 0$ the envelope $H(s) = Y e^{\beta s x_{00}} e^{|\beta s| X}$ equals $Y(1+s)^0 e^{\beta s(x_{00}+X)}$, which is exactly the shape `twoD_cutoff_analytic` asks for with $D = 0$, $L = x_{00} + X$, $C_0 = Y$.

3 Paper-facing meaning

`twoD_cutoff_amplitude` is the $d = 2$ equal-exponent instance of the τ-cutoff expansion of `thm:TaylorTree` under the paper's standing hypothesis that ξ and η are analytic on a neighbourhood of the closed box: analyticity on $|u|, |v| < \rho'$ with $\rho' > b$ gives weighted summability at any $\rho \in (b, \rho')$, and the theorem then produces the reduced face-and-corner expansion with an explicit remainder order and no compatibility, moment or envelope hypotheses. The exponent $L = x_{00} + X_\rho$ is a crude bound for $\sup \xi$ on the box, harmless for the Gaussian moments in s. Remaining paper-facing gaps for this block: grouping the three log families by exponent (collision-finset identity), the quantitative first-order constant, the noncritical parameter adapter, and general d.

4 Lean notes

- The whole unit compiled on the first attempt apart from one unused hypothesis and one long line: the API of units 117–118 (`wnorm_conv_le`, `wnorm_expCoeff_le`, `dblSum_conv`, `dblSum_expCoeff`) was designed to compose exactly here.
- Absolute summability of the evaluated family is `(dblSum_abs_le_wnorm ...).1.of_norm`; wrapping it as `dblSum_summable` keeps the additivity step (`Summable.tsum_add`) one line.
- A single weighted term is bounded by the weighted norm with `Summable.le_tsum` after unfolding `WSummable`; the side condition is discharged by `positivity`.
- `tsum_mul_left` needs no summability on $\mathbb{R}$; pulling the scalar $e^{\beta s x_{00}}$ out of the double sum is `rw [← tsum_mul_left]` plus a `tsum_congr ... ring`.